PROJECT PROPOSAL

AI-Driven Personalised Fitness Coach

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Course	Artificial Intelligence (AL-2002)
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1. Project Overview

● Project Topic:

This project proposes an AI-Driven Personalised Fitness Coach — an intelligent software system that generates, adapts, and optimises workout and nutrition plans for individual users based on their fitness goals, health data, and real-time feedback. The system leverages multiple AI paradigms including supervised learning for user profiling, genetic algorithms for workout plan optimisation, Bayesian networks for injury risk assessment, and unsupervised learning for fitness-level clustering. The coach evolves with the user over time, making it a genuinely adaptive and intelligent system rather than a static rule-based app.

● Objective:

The main goal is to develop an AI-powered fitness assistant that:

- Generates personalised weekly workout and diet plans based on user-inputted health data (age, weight, fitness level, goals)
- Optimises plans using a Genetic Algorithm to balance intensity, recovery, and progression
- Clusters users into fitness profiles using unsupervised learning (K-Means) to provide peer-benchmarked recommendations
- Predicts injury risk using a Bayesian Network built on user load and recovery data
- Adapts the plan week-over-week based on user performance feedback using a feedback loop
- Provides a clean, interactive interface for users to log workouts and receive updated recommendations

2. System Description

● Original Problem Background:

Most people who attempt to get fit rely on generic workout plans found online or expensive personal trainers. Generic plans ignore individual differences in fitness level, recovery capacity, body type, and schedule — leading to poor results, overtraining injuries, or early dropout. There is a clear gap for an intelligent, adaptive system that reasons about a user's unique data and continuously improves its recommendations. This is precisely the class of problem that modern AI techniques are designed to solve.

● Innovations Introduced:

- Genetic Algorithm for Plan Optimisation: Rather than producing a fixed workout plan, the system generates a population of candidate weekly plans and evolves them over generations — selecting, crossing over, and mutating plans based on a fitness function that rewards goal alignment, balanced muscle group coverage, and adequate rest. This directly applies the Genetic Algorithm.
- Bayesian Network for Injury Risk Prediction: The system models the probability of overuse injury as a function of training volume, sleep hours, soreness score, and consecutive training days — using a Dynamic Bayesian Network to flag high-risk weeks and automatically reduce load.
- Unsupervised Clustering for User Profiling: K-Means clustering groups users into fitness archetypes (e.g., beginner-sedentary, intermediate-active, advanced-athletic), enabling the system to benchmark a user against their peer group and provide contextually relevant recommendations
- Supervised Learning for Goal Prediction: A trained classifier predicts the most suitable plan category (weight loss, muscle gain, endurance, flexibility) from user input features — applying supervised learning
- Adaptive Feedback Loop: After each week, user-reported performance data (reps completed, perceived exertion, soreness) feeds back into the GA fitness function and Bayesian model, causing the plan to continuously evolve — making the coach genuinely intelligent over time.

3. AI Approach and Methodology

AI Technique	Course Connection	Role in System
Genetic Algorithm	Lab 6	Evolves and optimises weekly workout plans
K-Means Clustering	Lab 11	Groups users into fitness archetypes
Supervised Classification	Lab 11	Predicts optimal plan type from user profile
Bayesian Network / DBN	Lab 9	Models and predicts injury risk probability
EDA & Data Visualisation	Lab 10	Analyses user data trends and progress over time

Feedback / Adaptive Loop	Beyond syllabus	Continuously refines plans from user performance data

4. System Rules and Mechanics

● Modified Rules / System Workflow:

User Input → Profile Creation & Classification → K-Means Clustering → Genetic Algorithm Plan Generation → Bayesian Injury Risk Check → Plan Output → User Logs Weekly Performance → Feedback Loop Updates GA Fitness Function → New Optimised Plan Generated

● Output / Success Conditions:

- A personalised weekly workout plan (exercise name, sets, reps, rest periods) is generated for the user
- An injury risk score (Low / Medium / High) is displayed with explanatory factors
- User fitness profile cluster is identified and displayed with peer comparison stats
- Week-on-week plan improvement is measurable — plan adapts based on logged performance
- EDA dashboard shows progress graphs: volume over time, consistency rate, predicted goal timeline

● Process Sequence:

1. User registers and inputs profile: age, weight, height, fitness level (1-5), primary goal, available days per week, any injuries or limitations
2. Supervised classifier predicts the user's ideal plan category (e.g., Fat Loss, Hypertrophy, Endurance)
3. K-Means clusters the user into a fitness archetype; peer-benchmark stats are retrieved for that cluster
4. Genetic Algorithm initialises a population of 50 candidate 7-day workout plans; evolves for 100 generations using a fitness function scoring goal alignment, muscle balance, and recovery ratio
5. Best plan from GA is passed through the Bayesian Network — if injury risk is High, volume is automatically reduced by 20% and flagged to the user
6. Final plan is presented to the user via an interactive interface; user can accept or request a re-run
7. After each week, user logs completions and rates perceived exertion; this data updates the GA fitness function weights and Bayesian priors for the next week's plan

5. Libraries and Tools

Library / Tool	Purpose

Python 3.x	Core programming language for all AI components
NumPy / Pandas	Data handling, user profile storage, and numerical operations
Scikit-learn	K-Means clustering, supervised classification (Decision Tree / KNN), EDA utilities
pgmpy	Bayesian Network and Dynamic Bayesian Network construction and inference
Matplotlib / Seaborn	EDA visualisations, progress charts, cluster plots
Streamlit (or Tkinter)	Interactive user interface for profile input, plan display, and weekly logging
Random / Math (Python stdlib)	Core modules used in GA implementation (already covered in Week 2)

6. References

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